

the board is submerged completely. Use your paper towels to keep wiping the board while it is still submerged.

Doing this for a few minutes will dissolve the copper, leaving you with a board and a green etching solution. Remove the board and wash it in hydrogen peroxide. The remaining acid will be diluted.

Just to be sure that things have turned out the way you wanted, rinse your board under fresh water. If there is still some copper left, you can use your sandpaper to do away with it. However, this is a critical process and you should be careful about not scraping out the circuit that is present there.

Use nail lacquer as a final step to clean the sheet. This should complete the procedure entirely.

8. Drilling holes

This is a crucial step and one that requires fine craftsmanship. Here, a Dremel is absolutely necessary. You will be required to drill holes at all the connecting holes of the circuit. Use a 1/32" drill bit to do the work. Any other drill will not work. Also, drill holes at the corners so that they can be used to fixate the board somewhere.

9. Bringing everything together

Now comes the assembling. For this portion of your work, use a proper diagram that will guide you to do the same. Attach the wires and the other components as mentioned in the diagram.

You will most likely require your soldering tools here to do the work. After assembling everything together, what you will have is an Arduino that is ready to work. Your hardware part of the work is done.

10. Programming your Arduino

Now we shall think about the software bit. To make your Arduino work, you will need a breadboard and some more wires and components. A few resistors and power supply will also be of use here.

Connect the Arduino according to your program requirements and you are good to go!

Bar graph display and potentiometer

Bar graph displays are used to represent status indications, signal strength, and error indications. Instead of LEDs, the bar graph display and the potentiometer that supports it's working can be attached. Diagrams are